
ME 636: Combustion Technology

Course Code	ME 636
Course Name	Combustion Technology
L-T-P-C	3-0-0-3
Prerequisites	Instructor Consent
Intended for	UG/M. Tech. /MS/PhD
Distribution	Elective
Approval	10th Senate (Jun 2016)

Course Contents

- **Module I**

Introduction to combustion processes; combustion thermodynamics, reaction kinetics and combustion transport. Chain reactions, ignition, quenching and flammability limits, detonations, deflagrations and flame stability; Introduction to turbulent premixed combustion; Applications in IC engines, furnaces, gas turbines, and rocket engines. [4 Lectures]

- **Module II**

Fuels - Characteristics and Properties; Combustion Thermodynamics and Thermo-chemistry, Heat of Reaction, Calorific Value, Adiabatic Flame Temperature etc

Combustion Kinetics – Reaction Mechanism / Pathways, Rate Constants, Activation Energy; Flame –Diffusion Flame, Mixed Flame, Flame Velocity; Formation of Pollutants – CO, Soot, NOX and SOX; Combustion Modeling – Solid Combustion, Gas Combustion and Liquid. [12 Lectures]

- **Module III**

Combustion - Combustion Equipment: Combustion in Boilers (including Fluidized Bed Combustion), Liquid Combustion – Atomizer, Spray Combustion etc.; Gas Combustion – Gas Burners

Interchangeability of Fuels, Special Equipment; Engines, Combustion Phenomenon in Engines, Performance of Engine and Emissions; Stoves; Catalysis – Catalytic Combustion and Control of Emissions. [14 Lectures]

- **Module IV**

Advanced treatment of fundamental combustion processes; Conservation equations for reacting gas mixtures; The structure of one-dimensional diffusion and premixed flames; introduction to activation energy asymptotics. Two-dimensional Burke-Schumann flames and boundary layer combustion; Flame instabilities and flame stretch; turbulent combustion. [12 Lectures]

Text Books

1. Stephen Turns, **An Introduction to Combustion: Concepts and Applications**, 3rd Edition, McGraw Hill Education India Private Limited, 2012.
2. Mishra D P., **Fundamentals of Combustion**, PHI Learning Private Limited-New Delhi, 2010.